ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1
PAPER 1 November 2023

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Work out $(7 \times 10^3) \div 10$. Give the answer in standard form. **[2]**

2. Expand $(x + 1)(x + 5)$. **[3]**

3. $n(A) = 13$, $n(B) = 10$, $n(A \cap B) = 3$. Find $n(A \cup B)$. **[2]**

4. Simplify $(5^4)^3$. Leave as a power of 5. **[2]**

5. A map of Chitungwiza uses scale 2 : 500000. A road is 8 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm ↔ 1 km if 2:5 is first simplified — instead: scale 2 cm to 5 km.) **[3]**

6. Find the HCF and LCM of 24 and 42. **[3]**

7. Work out $3/5 \div 2/3$. **[3]**

8. Express 72 as a product of its prime factors. **[4]**

9. Evaluate $-5 - (-12)$. **[3]**

10. Express 24 as a percentage of 60. **[3]**

11. Factorise $x^2 + 8x + 15$. **[4]**

12. Solve<<
>>x + y = 8<<
>>x − y = -2. **[4]**

13. A trader at Unit L buys a sack of onions for $20 and sells for $32. Percentage profit? **[4]**

14. Factorise completely $24x + 32$. **[3]**

15. Write the equation of the line with gradient 2 passing through (0, -3). **[3]**

16. Solve $3(x + 3) = 30$. **[4]**

17. Make h the subject of $A = 2\pi r(r + h)$. **[4]**

18. In right-angled triangle PQR, $\hat{P} = 30°$ and hypotenuse PR = 10 cm. Find PQ, opposite 30°. **[4]**

19. In triangle ABC, $\hat{A} = 49°$ and $\hat{B} = 44°$. Find $\hat{C}$. **[3]**

20. Work out (3 3 ; 2 2)(2 ; 3). **[4]**

21. Circumference of a circle radius 14 cm. Take $\pi = 22/7$. **[4]**

22. A bag contains 3 red, 6 blue and 4 green beads. P(red)? **[3]**

23. Marks (frequency): 2 (3), 3 (5), 4 (6), 5 (2). Find the mean mark. **[4]**

24. A regular polygon has 10 sides. Find each interior angle. **[4]**

25. A sequence is 9, 11, 13, 15, … Find an expression for the nth term. **[4]**

26. A cuboid water tank measures 12 m by 4 m by 3 m. Find its volume. **[3]**

27. Find the midpoint of (3, 5) and (5, 8). **[3]**

28. Solve $3x + 3 < 12$. **[3]**

29. A right-angled triangle has shorter sides 12 cm and 16 cm. Find the hypotenuse. **[4]**

30. A triangular plot has base 6 m and perpendicular height 9 m. Find its area. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).